

Monodromy Integrality Barrier for Signed Moment Fits

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Abstract

After idempotent polynomial selectors collapse to binary masks, one may try arbitrary signed shell weights. On the expanded RH-254 windows, minimum-norm real weights fit the order-2–12 anchor tolerance at all 32 endpoints. Every fit is fractional, however, and maximum weights range from 39.64 to 3.02×10^{11} . We prove that the weighted moment germ equals $\prod_j (1 - \lambda_j z)^{w_j}$ near zero and has monodromy $e^{2\pi i w_j}$ around λ_j^{-1} . Integer combined exponents are therefore necessary for a single-valued meromorphic determinant quotient. The 32 continuous signed fits are finite compressions, not legal selectors.

1 Weighted moment germs

Let $\lambda_1, \dots, \lambda_J$ be distinct nonzero complex numbers and let $w_j \in \mathbb{C}$. Define the local logarithmic moment germ

$$\mathcal{F}_w(z) = \exp \left[- \sum_{n \geq 1} \frac{z^n}{n} \sum_{j=1}^J w_j \lambda_j^n \right] \quad (1)$$

for $|z| < \min_j |\lambda_j|^{-1}$.

Theorem 1.1 (Monodromy integrality). *On the initial disk,*

$$\mathcal{F}_w(z) = \prod_{j=1}^J (1 - \lambda_j z)^{w_j}. \quad (2)$$

Analytic continuation once around $z = \lambda_j^{-1}$ multiplies the germ by $e^{2\pi i w_j}$. Hence $\mathcal{F}_w$ is single-valued meromorphic near all reciprocal roots if and only if every combined exponent at a repeated root is an integer.

Proof. The power-series identity $-\sum_{n \geq 1} (\lambda z)^n / n = \log(1 - \lambda z)$ proves (2) on the initial disk. Continuing the logarithm once around its zero changes it by $2\pi i$; exponentiation therefore produces the stated factor. Trivial monodromy is equivalent to $e^{2\pi i w_j} = 1$, or $w_j \in \mathbb{Z}$. Integer powers are rational meromorphic factors, proving the converse. $\square$

Remark 1.2 (Necessary, not sufficient). Integer weights are necessary for a determinant ratio, but an operator realization also requires the corresponding algebraic multiplicities and invariant spaces. Negative integers describe denominator or superdeterminant contributions; they are not automatically present in the noisy operator.

2 Expanded-window signed fit

At each endpoint let V be the real order-2–12 shell-power matrix and d the difference between the Perron/parity-removed trace and deterministic anchor. We solve the unconstrained minimum-norm problem

$$\min\{\|w\|_2 : Vw \text{ is the least-squares approximation to } d\}. \quad (3)$$

This is deliberately more permissive than the RH-255 box [1].

quantity	value
endpoints within local tolerance	32/32
full row-rank systems	28/32
weighted residual range	2.15×10^{-15} –0.003783
fractional weights per endpoint	20–34
maximum absolute weight range	39.64– 3.02×10^{11}
integer-weight fits	0/32
maximum monodromy defect	1.9999999996

The four rank-deficient systems still meet their comparatively larger local tolerances. The smallest singular value across the batch is 2.71×10^{-19} , which helps explain the coefficient explosion. The fit is therefore both nonintegral and numerically ill-conditioned.

Corollary 2.1 (Scoped illegality of the continuous fits). *None of the 32 least-norm signed fits defines a single-valued finite determinant quotient with the fitted shell exponents.*

Proof. Every fitted vector contains at least 20 weights farther than 10^{-8} from the integer lattice. Apply Theorem 1.1. $\square$

3 Boundary and next route

This result does not exclude a bounded integer signed selector or a non-product quotient-kernel cancellation. It does exclude the inference “finite signed fit implies determinant selector.” RH-258 should audit a small integer lattice and then separately ask for an operator realization. Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, zeta-divisor equality, or RH implication is claimed.

References

- [1] Bin Wang. Expanded-window anchored zonotope obstruction. *Preprint*, 2026. RH-255 preprint.